Kunal Sharma

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PROFESSIONAL SUMMARY

Senior Data Engineer with nearly 7 years designing, building, and operating **large-scale data platforms, pipelines, and production AI systems** across financial services, banking, higher education, and telecom. Deep **Python, SQL, Spark, Snowflake, Databricks, dbt, Airflow, Kafka, Kubernetes, Terraform, and AWS**, with production **LLM/RAG systems on AWS Bedrock**, churn modeling across three industries, enterprise **governance and security**, and technical leadership as one of a few senior engineers. **AWS Certified Solutions Architect**.

KEY HIGHLIGHTS

- **Platform architecture at scale** — architected lakehouse, warehouse, and pipeline infrastructure at TB-to-petabyte scale, cutting runtimes 60% as architecture co-owner
- **Production GenAI** — LLM, RAG, and anomaly-detection systems on AWS Bedrock, cutting incident response 40%; personal AI digital-twin project deployed on Hugging Face
- **Churn modeling across three industries** — XGBoost/gradient-boosting models at Kent State (operationalized for outreach targeting), Cognizant (credit card), and Wipro (telecom retention)
- **Governance and security as defaults** — RBAC, dynamic masking, row access policies, tagging, lineage, PII/PCI, CyberArk, and SailPoint across regulated data
- **Dual-sided Alteryx** — Designer development and Server administration (EC2/ECS, MongoDB, RDS Aurora)
- **Technical leadership** — engineering standards, mentoring, cross-functional influence, and AI-assisted development adoption (Claude Code, Cursor, Copilot) as one of a few senior engineers

WORK EXPERIENCE

S&P Global **Jul 2023 – Present**
Senior Software Engineer II *Dallas, TX*

- **Architected and owned lakehouse and warehouse infrastructure end-to-end — Databricks** (Delta Lake, **Unity Catalog**, Databricks Apps), **Snowflake**, and **AWS** — cutting pipeline runtimes **60%** at TB-to-petabyte scale, as architecture co-owner and one of a few senior engineers.
- **Built ELT/ETL and streaming pipelines — dbt, Spark / PySpark, Python, Kafka, Kinesis,** and **CDC** (Debezium) — with **Airflow** orchestration, medallion architecture, and dimensional and semantic modeling.
- **Shipped production GenAI — LLM, RAG, and anomaly-detection systems on AWS Bedrock** (grounding datasets, chunking, embeddings, vector retrieval) plus **SageMaker** — cutting incident response **40%**.
- **Went deep on Snowflake-native engineering — Dynamic Tables, Streams and Tasks, Snowpipe Streaming, Snowpark, dynamic data masking, row access policies,** and **object tagging**.
- **Operated Kubernetes in production — EKS** with **custom controllers and operators**, autoscaling, and Docker; ran a quarter-plus **storage-side Kubernetes proof of concept** (Persistent Volumes, CSI drivers, storage classes) for data-layer throughput in a dev environment.
- **Ran dual-sided Alteryx — Designer** workflow development and **Server administration** on **EC2/ECS** with **MongoDB** and **RDS Aurora** (operation and performance tuning).
- **Owned governance and security — RBAC,** PII/PCI controls, **lineage and traceability** (Unity Catalog), **CyberArk** secrets management, and **SailPoint** entitlements in a regulated environment.
- **Built observability and data quality as engineering** — automated tests, **drift and anomaly detection, monitoring and alerting** (**Prometheus, Grafana, CloudWatch**) — not spot checks.
- **Automated infrastructure and delivery — Terraform** IaC, **CI/CD** (**GitHub Actions**), and **Trino / Starburst** (working) for federated query alongside **Athena**.
- **Worked across Azure — Azure Data Factory, Azure Databricks, Azure DevOps, Entra ID,** and **Key Vault** (working tier) alongside the AWS core.

- **Developed APIs and services** — **REST** (**Java / Spring Boot** microservices; **Python** with **FastAPI / Flask** at working tier) for data access and model integration.
- **Led without authority** — set engineering standards, mentored through code review and pairing, delivered **BI** (**Tableau, Power BI**), and drove **AI-assisted development** adoption (**Claude Code, Cursor, Copilot** — daily).

Cognizant **Aug 2022 – Jul 2023**

Senior Associate, Data Science Engineer *Dallas, TX*

- **Built churn-prediction models for credit card customers** — **XGBoost** and scikit-learn on multi-source **CRM** and transaction data — for banking client **Capital One**.
- **Owned the data engineering behind the models** — **Databricks / PySpark, Snowflake,** and **AWS** — cutting report time **35%**.
- **Built streaming and big-data pipelines** — **Kafka** real-time processing and **Hadoop / Hive** batch workloads.
- **Implemented CI/CD** — **Jenkins** pipelines with automated testing and code review.
- **Applied security controls** — RBAC and sensitive-data handling on financial data warehouses.
- **Consulted client-facing** — scoped ambiguous business problems, defined success criteria, and translated technical findings into executive-ready narratives.

Kent State University **Jan 2022 – Aug 2022**

Data Analyst *Kent, OH*

- **Built and operationalized a churn-risk model** — **XGBoost** predicting churn for **DPAE outgoing invitations**, with output driving real outreach targeting decisions and improving **marketing ROI**.
- **Ran marketing analytics end-to-end** — **attribution modeling** (multi-touch, last-click), **campaign ROI**, **customer segmentation**, and **A/B testing** on campaigns and messaging with significance testing.
- **Delivered FP&A** — **forecasting, budgeting, variance analysis,** headcount and spend planning, and FP&A dashboards for senior stakeholders.
- **Built ETL and data models** — **Python** and **Informatica** (working tier) from diverse sources into **IBM Cognos**, joining **marketing, CRM, and sales data** across the lifecycle.
- **Wrote advanced SQL** — CTEs, window functions, validation, and reconciliation for analysis-ready data.
- **Communicated insight** — **Tableau** and **Power BI** dashboards and presentations for non-technical partners.

Wipro Technologies **Aug 2018 – Dec 2020**

Project Data Engineer *Bangalore, India*

- **Built churn-prediction models for telecom customers** — **Random Forest and gradient boosting** in **Python** — that marketing acted on for retention campaigns.
- **Owned production on-prem pipelines end-to-end** — **Unix** environment (**Oracle SQL / PL/SQL, Python, Java, Shell, SSIS**) with POSIX/filesystem-level data handling.
- **Segmented customers and joined CRM data** — behavioral and operational data for campaign targeting.
- **Improved SLA/OLA compliance 50% and customer satisfaction 32%** — through reliable, automated, monitored operations with on-call ownership.
- **Tuned complex SQL and stored procedures** — query and pipeline performance for high-volume workloads.
- **Practiced disciplined engineering** — Git versioning, code reviews, Agile delivery, and documentation; early-career **Angular / TypeScript** web development during academic training.

PERSONAL PROJECTS

AI-Powered Digital Twin **2026**

Personal Project — deployed on Hugging Face

- **Built a conversational AI digital twin of myself** — an LLM application that answers as me, grounded in my own background and writing.
- **Implemented the full RAG pipeline** — document **chunking, embeddings,** and **vector stores** for retrieval-grounded responses.
- **Engineered prompts and context** — **prompt engineering, context management,** and **token and API cost management** for efficient, controlled generation.
- **Built LLM tool calling** — **parallel and sequential tool-calling** workflows for multi-step tasks.
- **Deployed publicly on Hugging Face** — live, shareable application.

EDUCATION

Kent State University **May 2022**
Master of Science, Business Analytics

APJ Abdul Kalam Technical University **May 2018**
Bachelor of Technology, Computer Science and Engineering

CERTIFICATIONS AND SKILLS

- **Certifications:** AWS Certified Solutions Architect - Associate
- **Languages: Python** (expert), **SQL** (expert — CTEs, window functions, temporal data, tuning), **PL/SQL, T-SQL**, **Java** (Spring Boot), **PySpark**, **Shell**; TypeScript / JavaScript (working — Angular, academic background), Scala (familiar)
- **Data Warehouses and Lakehouses: Snowflake** (deep — Dynamic Tables, Streams and Tasks, Snowpipe Streaming, Snowpark, masking, row access policies, tagging), **Databricks** (Delta Lake, Unity Catalog, Databricks Apps), **Redshift**, **PostgreSQL**, Oracle, SQL Server, **RDS Aurora** (operation and tuning), MongoDB, DynamoDB; BigQuery, ClickHouse, Neo4j, Spanner, Elasticsearch (familiar)
- **ETL/ELT and Orchestration: dbt**, **Apache Airflow**, **AWS Glue**, **SSIS**, **Informatica** (working); Talend, Dagster (familiar); medallion architecture, dimensional and semantic modeling, Kimball, schema evolution
- **Streaming and Query Engines: Apache Kafka**, **AWS Kinesis**, **CDC** (Debezium), **Trino / Starburst** (working), **Athena**; Apache Flink, Apache Beam, Iceberg (familiar)
- **Big Data: Apache Spark / PySpark** (deep), **Hadoop / Hive**, TB-to-petabyte scale, Parquet and Delta formats, unstructured-to-structured transformation
- **AWS: S3, EC2, ECS, Lambda, Glue, EMR, Athena, Redshift, Kinesis, Step Functions, Bedrock, SageMaker, CloudWatch, RDS Aurora** — certified Solutions Architect
- **Azure and GCP: Azure** (working) — **Data Factory, Azure Databricks, Azure DevOps, Entra ID, Key Vault**; **GCP** (familiar) — BigQuery, Dataflow, Pub/Sub, Cloud Functions, Vertex AI
- **Kubernetes and Infrastructure: Kubernetes / EKS** (custom controllers, operators, autoscaling), **storage-side Kubernetes** — Persistent Volumes, CSI drivers, storage classes (proof of concept), **Docker**, **Terraform**, **CI/CD** (GitHub Actions, Jenkins, Azure DevOps)
- **Observability and Quality: Prometheus, Grafana, CloudWatch** (Datadog familiar), automated data-quality testing, **drift and anomaly detection**, alerting, SLA ownership, on-call
- **Governance and Security: RBAC**, **dynamic data masking, row access policies, object tagging**, **lineage and traceability** (Unity Catalog), PII/PCI, **CyberArk** (secrets), **SailPoint** (entitlements), data classification, metadata management, quality rules
- **Machine Learning: Churn and propensity modeling** (three industries — education, credit card, telecom), **XGBoost, gradient boosting, LightGBM**, scikit-learn, Random Forest, regression, SVM, clustering, **segmentation, forecasting**, Pandas, NumPy, **SageMaker**; TensorFlow, PyTorch, MLFlow (familiar); feature stores (project)
- **GenAI and LLMs: Production LLM systems** (**AWS Bedrock** — RAG, anomaly detection, decisioning), **RAG engineering** (**chunking, embeddings, vector stores**), **prompt engineering, context management, token and API cost management**, **LLM tool calling** (parallel and sequential), Hugging Face deployment; agentic and multi-agent workflows, LangChain, LlamaIndex (project); MCP servers (new)
- **APIs and Services: REST API** design and development (**Java / Spring Boot** microservices, production; **Python — FastAPI, Flask**, working), API consumption and third-party integration, event-driven architecture
- **BI and Analytics: Tableau, Power BI, Alteryx** (**Designer development + Server administration**), IBM Cognos, Excel (advanced), **A/B testing, significance testing, causal inference**, **attribution modeling, campaign ROI, FP&A** (forecasting, budgeting, variance)
- **Leadership and Practice: Technical leadership** (standards, architecture co-ownership, independent technology decisions), **mentoring** (code review, pairing, design sessions), cross-functional and executive communication, **project leadership** (requirements through deployment), Agile/Scrum, **AI-assisted development** (**Claude Code, Cursor, Copilot** — daily, with production guardrails)